Pendem Nitesh Sri Satya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	5.98
SASI Educational Institute, Velivennu	Class XII (CBSE)	2024	93.80%
SASI Educational Institute, Nidadavole	Class X (SSC)	2022	94.00%

SKILLS

- **Programming Languages:** C, Python, SQL
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, PyTorch, Tensorflow, Keras, Streamlit, NLTK
- **Version Control:** Git, GitHub
- **Operating Systems:** Linux, Windows
- **Development Tools:** Jupyter Notebook, Google Colab, Visual Studio Code
- **Core Concepts:** Data Structures and Algorithms, Object-Oriented Programming (OOPS), Database Management System (DBMS), Machine Learning, Deep Learning
- **Soft Skills:** Communication, Team Collaboration, Problem Solving, Time Management

RELEVANT COURSES

- **Artificial Intelligence:** Foundations of Machine Learning, Deep Learning, Introduction to Modern AI, AI for Humanity, Programming for Artificial Intelligence
- **Computer Science:** Introduction to Programming, Data Structures and Algorithms, Database Management System
- **Mathematics:** Discrete Mathematics, Calculus, Metric Spaces, Optimization, Probability and Random Variables

PROJECTS

- **Long Context Modelling using State Space Models (SSM)**
- Implemented state space models for efficient long-context sequence modeling and sentiment classification.
- Designed modular training pipelines with configurable optimizers and evaluated long-range dependency modeling through extensive experiments.
- **Transformer From Scratch**
- Built a complete transformer encoder implementing multi-head self-attention, positional encoding, residual connections, layer normalization, and feed-forward networks.
- Developed configurable training and inference pipelines for sequence modeling with modular architecture components.
- **Convolutional Neural Network From Scratch**
- Implemented a complete CNN framework from scratch including convolution, pooling, activation, etc..
- Trained and evaluated image classification models while analyzing feature maps and comparing optimization algorithms.
- **Heart Disease Classification**
- Developed an end-to-end machine learning classification pipeline including preprocessing, feature engineering, model selection, and evaluation.
- Compared multiple supervised learning algorithms using cross-validation and classification performance metrics.
- **Diamond Price Regression**
- Built a complete regression pipeline for diamond price prediction using feature engineering, preprocessing.
- Evaluated regression models using MAE, RMSE, and R^2 score tuning for improved prediction accuracy.
- **Sentiment Analysis using Recurrent Neural Network (RNN)**
- Implemented a simple RNN for IMDb sentiment classification using trainable word embeddings and sequence modeling.
- Performed text preprocessing, tokenization, sequence padding, and evaluated binary classification performance.
- **Next Word Prediction using LSTM**
- Developed an LSTM based language model for next-word prediction using embedding layers and sequential learning.
- Implemented vocabulary generation, sequence creation, and inference for real-time text generation.
- **Customer Churn Classification using Artificial Neural Network (ANN)**
- Built an Artificial neural network for customer churn prediction with feature encoding, scaling, and model optimization.
- Deployed the trained TensorFlow model through a streamlit application for real-time churn prediction.

EXTRACURRICULAR

- Actively participated in the **National Service Scheme (NSS)**, contributing to **community service** and various **social welfare initiatives**.

POINTS OF RESPONSIBILITY

Coordinator - Hospitality & Logistics | SPIC MACAY, IIT Hyderabad

- Coordinated hospitality and logistics for cultural events, ensuring seamless guest experience and event execution.
- Managed accommodation, transportation, and venue readiness for artists and distinguished guests.
- Collaborated with student volunteers and organizing teams to execute events within scheduled timelines.
- Resolved operational challenges during live events through effective communication and coordination.
- Supported end-to-end event management, contributing to the successful conduct of multiple SPIC MACAY programs.